

Blackout (grade 6 edition) — play-through review

Theme `blackout_ms` · middle-school physical science, grade 6: circuits, energy and the grid · 10 days, 33 stops · reviewed 2026-08-21 by reading `books/blackout-ms.yml` in full, working every board, and comparing against `blackout` and `blackout_fable`.

Verdict

The junior edition with the most arithmetic in it — nine of thirty-three stops are estimate boards, the highest count in the junior set — and all nine are correct and correctly pitched:

Board	Arithmetic
How much went missing	$6,400 - 6,100 = 300 \text{ MW}$
The power that never arrives	$1,150 \times 1.5 \text{ per hundred} \approx 17 \text{ MW}$
Why the far end sits low	$1.07 \div 11 \approx 9.7\%$
Power for four hours	$5.6 \times 4 = 22.4 \text{ MWh}$
What comes out against what went in	$187 \div 220 = 0.85$
What a short circuit is limited by	$6,350 \div 0.42 \approx 15,000 \text{ A}$
The peak, and what the cold adds	$6,400 + 120 \times 4 = 6,880 \text{ MW}$
What the dark hours cost	$300 \times 2.5 = 750 \text{ MWh}$
The same line, busier	$1,280 \times 1.6 \text{ per hundred} \approx 20 \text{ MW}$

Two of those are worth naming as good grade-6 writing. **"The power that never arrives"** and **"The same line, busier"** are the same calculation at two loads — 17 MW against 20 MW — so the player discovers by arithmetic that a 11% busier line loses 20% more, which is the square in I^2R without the square being mentioned. And the percentage boards are phrased as *"eleven and a half hundreds, and 1.5 lost from each hundred"*, which is how a sixth grader can actually do a percentage in their head.

The campaign's spine survives the derivation: four million people, one number, a corridor sensor that was confident and wrong, and a fortnight of holding two sides together. Chinelo Obi is named on the opening card with her job attached and her position stated — *"would rather switch off fourteen thousand homes than let the number fall"* — which is better than the fable edition manages.

Answerable: 33/33. **Sense:** Strong. The number-that-is-a-balance framing on the opening card is the clearest statement of grid frequency at any level in this repo. **Level:** Right. `questionLoad` reports 5 of 33 demanding stops (15%). **Fun:** High. "A fault takes seconds. Getting the power back takes days."

Implemented since this review

- **BM-01a is done and it went the other way from expected.** The general fix — an equation's own content words as keywords where its signature is unmatchable — was written, measured at **zero effect**, and reverted. What did work is one keyword: `the share` , added to fourteen lists, which makes the gate see "*Lost = what it carries × the share lost*" as `part = whole × share` . One row cleared, verified by hand against the board, with a before/after diff across all 42 themes confirming nothing else moved. **blackout_ms is 2 of 5 rather than 1 of 5**, and the other three rows are real.

Everything else in the findings table below is unchanged, and `npm run check` is green on this theme.

Findings

ID	Sev	Where	Issue	Proposed fix
BM-01	RECORDED	<code>engine/dev/curriculum-debt.json</code> , 4 rows — and the gate that produced them	The file records four of five syllabus relations as uncomputed: <code>how long = amount ÷ rate</code> , <code>part = whole × share</code> , <code>average = total ÷ how many</code> , <code>spare = what can be made – what is being used</code> . At least two are false. <code>part = whole × share</code> is computed, twice, by "The power that never arrives" and "The same line, busier", whose relationship reads " <i>Lost = what it carries × the share lost</i> " — <code>whole × share</code> , in the campaign's own words. The gate cannot see it because the equation has no symbols, so <code>symbolSignature</code> contributes nothing and everything rests on a five-word keyword list (<code>per cent</code> , <code>percent</code> , <code>percentage</code> , <code>share of</code> , <code>out of a hundred</code>) that does not contain the words the board actually uses. <code>spare = ...</code> is arguably also computed, with the sign flipped, by "How much went missing". CLAUDE.md's own rule applies: a debt file recording gaps that are not there is worse than no debt file , because the work list is wrong and the two real gaps hide among the false ones.	Measured against the boards: of the four rows, <code>part = whole × share</code> was false and is cleared. The other three are real — no board computes <code>how long = amount ÷ rate</code> or <code>average = total ÷ how many</code> , and <code>spare = capacity – demand</code> is genuinely a different quantity from the shortfall "How much went missing" computes (6,400 – 6,100). Each wants one written stop, and this edition has no three-stop day free to put one on, so it is a day-count change as well. Recorded.
BM-02	WORTH	Days 3, 8 and 10	Three of ten days author 4 stops, so those three carry no callback. With zero <code>–</code> Review variants (BM-03) the point is moot today, but it will bind the moment variants exist.	Prefer a move to a delete.

BM- 03	WORTH	Zero – Review variants	No spaced retrieval. Cross-campaign §9.	Three or four variants.
BM- 04	WORTH	engine/dev/concept-debt.json , 7 rows	Seven ordering rows and no takesAsRead available at grade 6, so all seven are ordering or writing. Two point at the same base — <i>current is the flow, voltage is the push and something in the way makes it harder for current to flow</i> both precede the breaker concept — which suggests one early stop claiming both would clear several rows at once.	Write the one claim rather than moving three stops. CLAUDE.md records that two batches of concept work made the total worse, so re-measure after each.
BM- 05	TASTE	Numeral damage, 3 occurrences	"1 million", "1 place". Minimal — this book is among the cleanest. Note that the parent has the same three and the fable edition three more.	Editorial pass. Cross-campaign §1.

What the derivation did well

- **It kept the number that is a balance.** "That rate is not really a reading. It is a balance: how much power the stations make against how much everyone is using." That is the parent's physics in a sixth grader's sentence.
- **Two boards that are one calculation at two loads**, so the square in I^2R arrives as a discovery rather than a formula.
- **The opening names a person with a position**, which two of the three Blackout editions do not.

Opening and closing

Opening: 111 words in nine short sentences, all four beats, and Chinelo Obi's position stated rather than her title recited. Good.

Closing: three paragraphs. Inherits the parent's, including the best unfinished-business line in the set — "the practice that would have caught all of it ... is a paragraph in a report until somebody on nights makes it a habit."

Warm-ups

Authored for this edition. No findings.